

# Deep Generative Models

## Lecture 10

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# Recap of Previous Lecture

## Continuous-Time Dynamics

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\psi_t(\mathbf{x}_0) = \int_{t_0}^t \mathbf{v}_{\theta}(\mathbf{x}(s), s) ds + \mathbf{x}_0 \approx \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0, t).$$

- ▶  $\mathbf{v}_{\theta} : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$  is a vector field.
- ▶  $\psi : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$  is a flow (the solution of ODE):

## Euler ODESolve

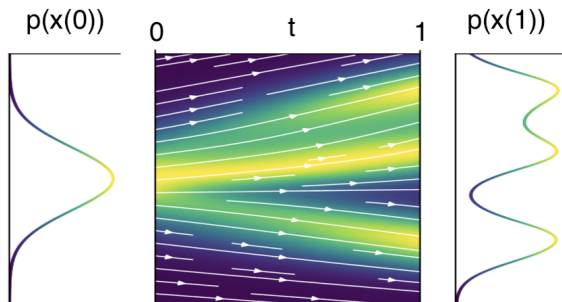
$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

More advanced numerical methods (such as Runge-Kutta) are often used instead of unstable Euler update step.

## Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_\theta(\mathbf{x}(t), t); \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

- ▶ Suppose  $\mathbf{x}(0)$  is a random variable with density  $p_0(\mathbf{x})$ . Then,  $\mathbf{x}(t)$  is a random variable with density  $p_t(\mathbf{x})$ .
- ▶  $p_t(\mathbf{x}) = p(\mathbf{x}, t)$  describes the **probability path** between  $p_0(\mathbf{x})$  and  $p_1(\mathbf{x})$ .



# Recap of Previous Lecture

## Theorem (Picard)

If  $\mathbf{v}$  is continuously differentiable with a bounded derivative in  $\mathbf{x}$  and continuous in  $t$ , then the ODE has a **unique solution** given by a flow  $\psi_t$ .

This guarantees the ODE is **uniquely reversible**.

$$\psi_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_\theta(\psi_t(\mathbf{x}_0), t) dt$$

$$\mathbf{x}(1) = \mathbf{x}(0) + \int_0^1 \mathbf{v}_\theta(\mathbf{x}(t), t) dt$$

$$\mathbf{x}(0) = \mathbf{x}(1) + \int_1^0 \mathbf{v}_\theta(\mathbf{x}(t), t) dt$$

**Note:** Unlike discrete-time flows,  $\mathbf{v}$  need not be invertible (uniqueness ensures bijection).

How can we compute  $p_t(\mathbf{x})$  at arbitrary  $t$ ?

## Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

### Theorem (Continuity Equation)

If  $\mathbf{v}$  is continuously differentiable in  $\mathbf{x}$  (with bounded derivative) and continuous in  $t$ , and  $p_t(\mathbf{x}) > 0$ , then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

### Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

- ▶ This solution gives us the density **along the trajectory**.
- ▶ However, it's difficult to efficiently estimate **the last term**.

# Recap of Previous Lecture

## SDE Basics

Let's define a stochastic process  $\mathbf{x}(t)$  with initial condition  $\mathbf{x}(0) \sim p_0(\mathbf{x})$ :

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w},$$

where  $\mathbf{w}(t)$  is the standard Wiener process (Brownian motion):

$$\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t - s)\mathbf{I}), \quad d\mathbf{w} = \epsilon \cdot \sqrt{dt}, \text{ where } \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

## Discretizing the SDE (Euler-Maruyama Update) – SDEsolve

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \epsilon \cdot \sqrt{dt}$$

- ▶ At each time  $t$ , we have the density  $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ .
- ▶  $p : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}_+$  is a **probability path** between  $p_0(\mathbf{x})$  and  $p_1(\mathbf{x})$ .

# Recap of Previous Lecture

## Theorem (Kolmogorov-Fokker-Planck)

If  $p_t(\mathbf{x}) \in C^{1,2}$  (i.e.,  $C^1$  in  $t$  and  $C^2$  in  $\mathbf{x}$ ), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

## Langevin SDE (Special Case)

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

The density  $p_t(\mathbf{x})$  is a **stationary** distribution for the SDE. Setting the stationary density to the model distribution,  $p_t(\mathbf{x}) = p_{\theta}(\mathbf{x})$  for all  $t$ :

## Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

# Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
3. Reverse SDE
4. Score-Based Generative Models Through SDEs
5. Flow Matching (FM)



# Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
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# Score Matching SDE

## Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t,$$

$$q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \boldsymbol{\epsilon}_{t-1},$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

# Score Matching SDE

## Denoising Score Matching

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$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \epsilon, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

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Let's transform this Markov chain into the continuous stochastic process  $\mathbf{x}(t)$  by letting  $T \rightarrow \infty$ :

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon}$$

# Score Matching SDE

## Denoising Score Matching

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$$\begin{aligned} \mathbf{x}(t) &= \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon} \\ &= \mathbf{x}(t - dt) + \sqrt{\frac{\sigma^2(t) - \sigma^2(t - dt)}{dt}} dt \cdot \boldsymbol{\epsilon} \end{aligned}$$

# Score Matching SDE

## Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

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## Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

## Score Matching SDE

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## Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$  is a monotonically increasing function.



## Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
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## Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$  is a monotonically increasing function.

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

# Diffusion SDE

## Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

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Let's turn this Markov chain into a continuous stochastic process by letting  $T \rightarrow \infty$  and setting  $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$  (where  $dt = \frac{1}{T}$ ):

$$\mathbf{x}(t) = \sqrt{1 - \beta(t)dt} \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon$$

# Diffusion SDE

## Denoising Diffusion

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$$\begin{aligned} \mathbf{x}(t) &= \sqrt{1 - \beta(t)dt} \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon \approx \\ &\approx \left(1 - \frac{1}{2}\beta(t)dt\right) \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon \end{aligned}$$

# Diffusion SDE

## Denoising Diffusion

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# Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

# Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

## Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as  $\mathbf{x}(0)$  has unit variance.

# Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$



# Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

## Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

## Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

We treat the discrete-time methods (NCSN, DDPM) as a special case of the continuous-time methods (VE-SDE, VP-SDE).

# Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
3. Reverse SDE
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# Probability Flow ODE

## ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) \Leftrightarrow \frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x}))$$

The only source of randomness is the initial distribution  $p_0(\mathbf{x})$ .

# Probability Flow ODE

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The only source of randomness is the initial distribution  $p_0(\mathbf{x})$ .

## SDE and KFP Equation

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

Now there are two sources of randomness: the initial distribution  $p_0(\mathbf{x})$  and the Wiener process  $\mathbf{w}(t)$ .

# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$  with  $p_t(\mathbf{x}) > 0$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

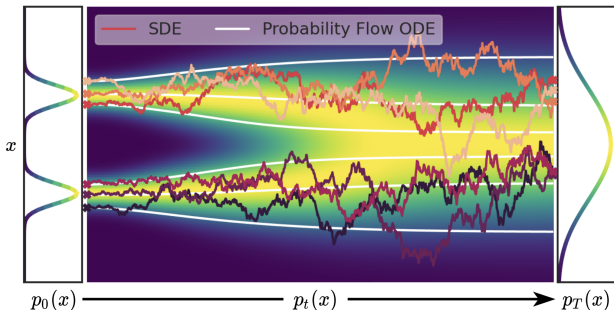
$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$  with  $p_t(\mathbf{x}) > 0$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

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Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

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## Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right)$$

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$$\begin{aligned} \frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = \\ &= \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} \right] \right) \end{aligned}$$



# Probability Flow ODE

## Theorem

Suppose the SDE  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  induces the probability path  $p_t(\mathbf{x})$  with  $p_t(\mathbf{x}) > 0$ . Then, there exists an ODE with the same probability path  $p_t(\mathbf{x})$ , given by

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# Probability Flow ODE

## Proof (Continued)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \text{tr} \left( -\frac{\partial}{\partial \mathbf{x}} \left[ \left( \mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right)$$

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# Probability Flow ODE

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$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}); \quad \tilde{g}(t) = 0$$

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- $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$  is the continuous-time score function.

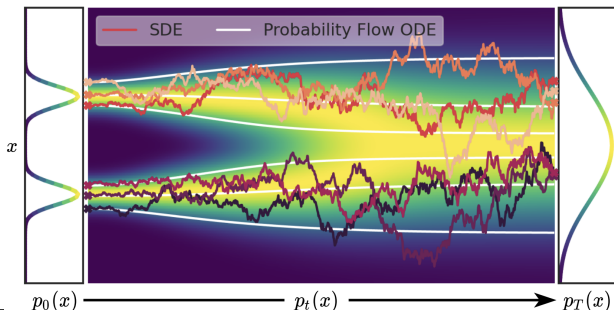
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- ▶  $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$  is the continuous-time score function.
- ▶ The ODE produces more stable trajectories.



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020



# Why convert SDEs to PF-ODEs? (1) Efficient Sampling

- ▶ Brownian increments scale as  $d\mathbf{w} \sim \sqrt{dt}$  — SDE trajectories are nowhere differentiable.
- ▶ SDEsolve (Euler–Maruyama): local error  $\mathcal{O}(\sqrt{\Delta t})$ .
- ▶ Higher-order SDE schemes need Lévy areas  $\Rightarrow$  impractical in high dimensions.

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PF-ODE drift  $\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)$  is smooth

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- ▶ Adaptive step-size control is standard for ODEsolve.
- ▶ In practice: 100–1000 SDEsolve steps  $\rightarrow$  20–50 ODEsolve steps.
- ▶ Caveat: SDE sampling can still win at high NFE — injected noise corrects accumulated error.

## Why convert SDEs to PF-ODEs? (2) Deterministic Generation

- ▶ The PF-ODE defines a deterministic, invertible map  $\mathbf{x}(0) \leftrightarrow \mathbf{x}(T)$ .
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### Exact log-likelihood (instantaneous change of variables)

$$\log p_0(\mathbf{x}(0)) = \log p_T(\mathbf{x}(T)) + \int_0^T \text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt$$

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- ▶ Connects PF-ODE to continuous normalizing flows.
- ▶ Invertibility enables image  $\rightarrow$  latent  $\rightarrow$  edited image workflows.
- ▶ Smooth interpolation in latent space between two samples.

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Same trained score  $\mathbf{s}_\theta(\mathbf{x}, t)$ ; only the discretization changes (reverse SDE derived in the next section).

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1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
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# Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
3. Reverse SDE
4. Score-Based Generative Models Through SDEs
5. Flow Matching (FM)

## Reverse SDE

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Here  $dt$  can be  $> 0$  or  $< 0$ .

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## Theorem

There exists a reverse SDE for  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  (assume  $p_t(\mathbf{x}) > 0$ ), given by:

$$d\mathbf{x} = \left( \mathbf{f}(\mathbf{x}, t) - g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt + g(t)d\bar{\mathbf{w}}$$

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*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

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- ▶ Since  $\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)$  (PF-ODE), the reverse drift equals  $\mathbf{f} - g^2\mathbf{s} = \mathbf{v} - \frac{1}{2}g^2\mathbf{s}$ .



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## Proof Sketch

- ▶ Convert the initial SDE to a probability flow ODE.
- ▶ Reverse the probability flow ODE.
- ▶ Convert the reversed probability flow ODE back to an SDE.

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*Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

# Proof of Reverse SDE

- Convert the initial SDE to a probability flow ODE:

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$$d\mathbf{x} = \left( -\mathbf{f}(\mathbf{x}, 1 - \tau) + \frac{1}{2}g^2(1 - \tau)\frac{\partial}{\partial \mathbf{x}} \log p_{1-\tau}(\mathbf{x}) \right) d\tau$$

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Here  $\mathbf{w}(\tau)$  is a Wiener process in forward  $\tau$ -time.

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## Theorem

There exists a reverse SDE for  $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$  (assume  $p_t(\mathbf{x}) > 0$ ), given by:

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Here  $d\tau > 0$ ,  $dt < 0$ , and  $\bar{\mathbf{w}}$  is reverse-time Wiener process.

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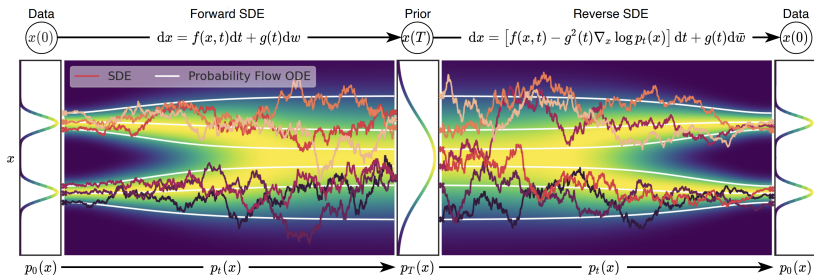
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## Score Matching SDE

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# Diffusion SDE

## Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

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# Outline

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# Score-Based Generative Models Through SDEs

## Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Is it possible to train score-based diffusion models in continuous time?



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# Score-Based Generative Models Through SDEs

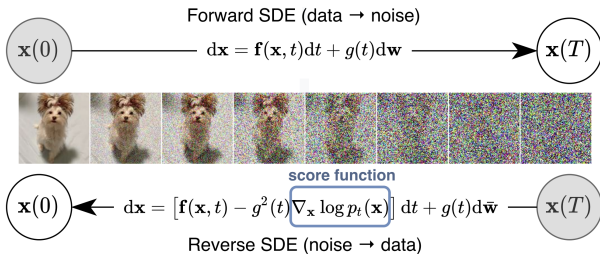
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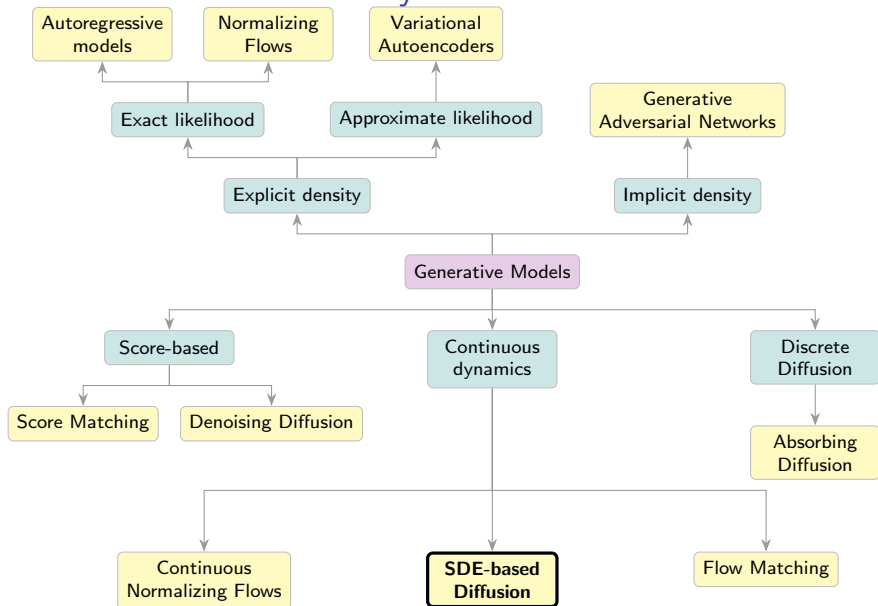
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Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

# Generative Models Taxonomy



# Score-Based Generative Models Through SDEs

## Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

# Score-Based Generative Models Through SDEs

## Continuous-Time Objective

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$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\right)$$

$$\nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) = -\frac{1}{\sigma^2(t)}(\mathbf{x}(t) - \boldsymbol{\mu})$$

**Note:** Normality holds for  $\mathbf{f}(\mathbf{x}, t)$  affine in  $\mathbf{x}$  ( $\mathbf{f}(\mathbf{x}, t) = f(t)\mathbf{x}$ ).

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$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w} \quad (\text{Variance Exploding SDE})$$

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w} \quad (\text{Variance Preserving SDE})$$

Is it possible to explicitly derive  $\boldsymbol{\mu}(\mathbf{x}(0), t)$  and  $\sigma^2(t)$  for VE-SDE and VP-SDE?

# Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\right)$$

## Theorem

The moments of the SDE  $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$  satisfy:

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)$$

$$\frac{d\sigma^2(t)}{dt} = 2f(t)\sigma^2(t) + g^2(t)$$

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## Proof

$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = \mathbb{E}[f(t)\mathbf{x}dt|\mathbf{x}(0)] + \mathbb{E}[g(t)d\mathbf{w}|\mathbf{x}(0)]$$



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# Score-Based Generative Models Through SDEs

## Theorem

$$\frac{d\mu(\mathbf{x}(0), t)}{dt} = f(t)\mu(\mathbf{x}(0), t)$$

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## Examples

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$$\mu = \mathbf{x}(0) \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right)$$



# Score-Based Generative Models Through SDEs

## Training

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2. Sample  $\mathbf{x}(t) \sim q(\mathbf{x}(t)|\mathbf{x}(0))$ .
3. Compute loss  $\mathcal{L} = \|\mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0))\|_2^2$ .

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## NCSN

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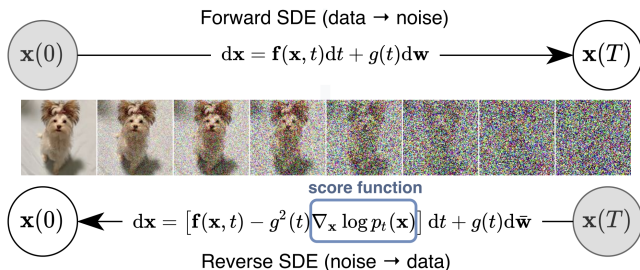
## DDPM

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0)e^{-\frac{1}{2}\int_0^t \beta(s)ds}, \left(1 - e^{-\int_0^t \beta(s)ds}\right) \cdot \mathbf{I}\right)$$

# Score-Based Generative Models Through SDEs

## Sampling

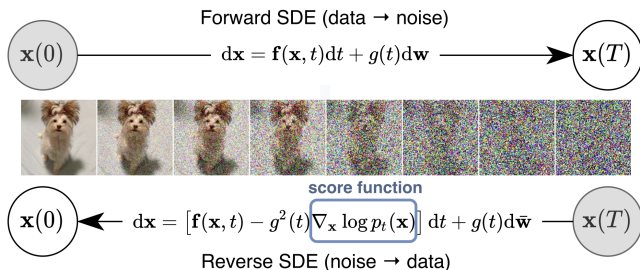
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2. Solve the reverse SDE using numerical solvers (SDEsolve).



# Score-Based Generative Models Through SDEs

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- ▶ Discretizing the reverse SDE provides ancestral sampling.
- ▶ Discretizing the probability flow ODE yields deterministic sampling.

Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

# Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
3. Reverse SDE
4. Score-Based Generative Models Through SDEs
5. Flow Matching (FM)

# Continuous-Time Normalizing Flows

Let's return to ODE dynamics  $\mathbf{x}_t = \mathbf{x}(t)$  in the interval  $t \in [0, 1]$ :

- ▶  $\mathbf{x}_0 \sim p_0(\mathbf{x})$ , where  $p_0(\mathbf{x})$  is a base distribution (mostly  $\mathcal{N}(0, \mathbf{I})$ );
- ▶  $\mathbf{x}_1 \sim p_1(\mathbf{x})$ , where  $p_1(\mathbf{x})$  is the true data distribution  $p_{\text{data}}(\mathbf{x})$ .

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

The velocity  $\mathbf{v}$  determines a flow  $\psi_t$ :  $\mathbf{x}_t = \psi_t(\mathbf{x}_0)$ .

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### KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left( \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$



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- ▶ It's hard to solve the continuity equation directly due to the trace term.
- ▶ There's a method (the adjoint method) that solves this equation directly, but it's unstable and unscalable.

# Continuous-Time Normalizing Flows

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- ▶ Knowing the vector field  $\mathbf{v}(\mathbf{x}, t)$ , the KFP (or continuity) equation allows us to compute the density  $p_t(\mathbf{x})$ .
- ▶ Flow matching provides scalable approach to Neural ODEs.

# Continuous-Time Normalizing Flows

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## Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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- ▶ Approximate the true vector field  $\mathbf{v}(\mathbf{x}, t)$  using  $\mathbf{v}_\theta(\mathbf{x}, t)$ .
- ▶ Use the learned flow  $\psi_t$  for deterministic sampling:  
 $\mathbf{x}_1 = \psi_1(\mathbf{x}_0)$ .

# Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

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$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

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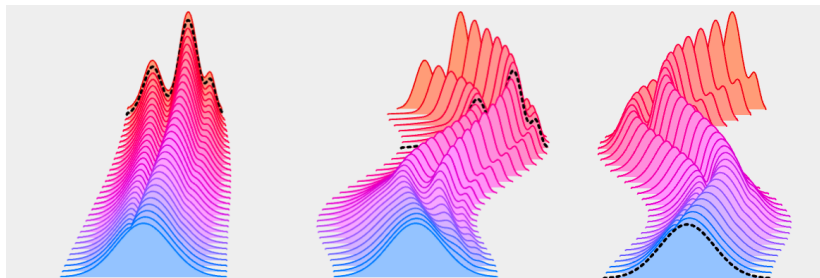
**Problem:** The true vector field  $\mathbf{v}(\mathbf{x}, t)$  is **unknown**.



# Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

- ▶ There are infinitely many possible  $\mathbf{v}(\mathbf{x}, t)$  between  $p_0(\mathbf{x})$  and  $p_1(\mathbf{x})$ .
- ▶ We need to select the "best"  $\mathbf{v}(\mathbf{x}, t)$  and make the objective tractable.



## Summary

- ▶ Score matching (NCSN) and diffusion models (DDPM) are discretizations of SDEs (variance exploding and variance preserving).
- ▶ Every SDE admits a corresponding probability flow ODE that follows the same probability path  $p_t(\mathbf{x})$ , yielding deterministic invertible sampling and exact log-likelihoods via the instantaneous change of variables.
- ▶ Converting SDEs to PF-ODEs enables efficient high-order numerical solvers, reducing the number of inference steps from 100–1000 to 20–50.
- ▶ SDEs can be reversed in time using the score function  $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ ; discretizing the reverse SDE gives ancestral sampling, while discretizing the PF-ODE gives deterministic sampling.
- ▶ The continuous-time score matching objective generalizes DDPM and NCSN; for affine-drift SDEs the transition kernel  $q(\mathbf{x}(t)|\mathbf{x}(0))$  is Gaussian, making the training objective tractable.